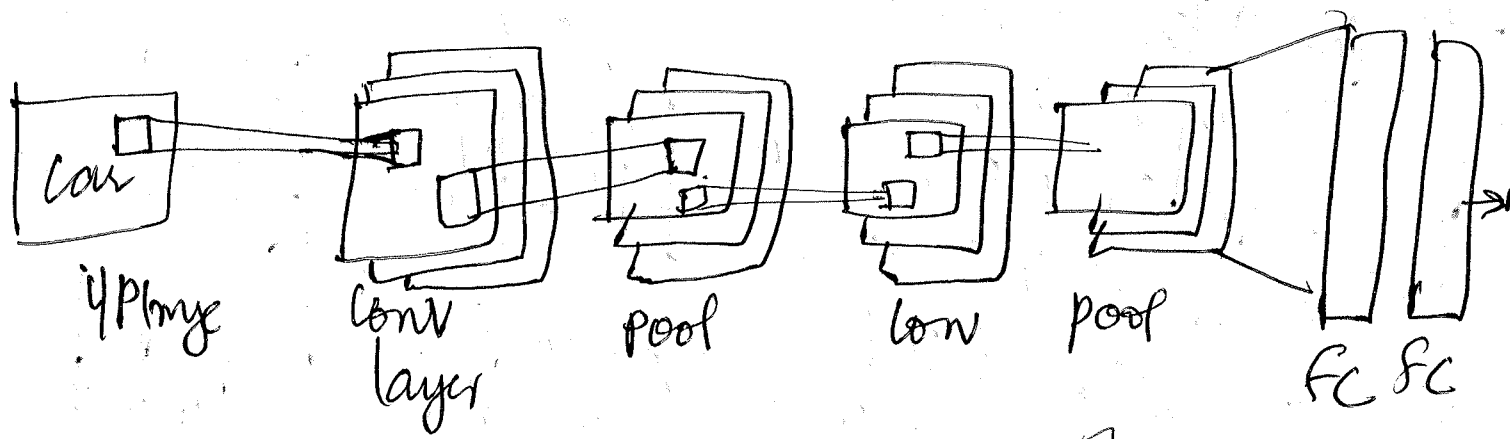


CNN



In CNN Conv ^① → pooling ^②
 ~~~~~  
 repeats

Softmax  
 (0 or 1)

- ③ fully connected layer (also cascade)
- ④ Softmax layer which do classification task

NN → we were feeding 1D image

CNN → 4 ANN → simple feed forward nn

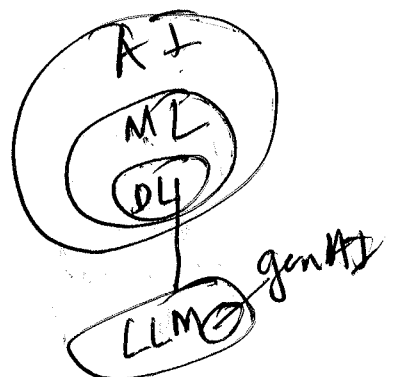
① convolut<sup>n</sup> if correlat<sup>n</sup> (no filter) flip  
 → flip the kernel

Conv (filter size, stride, padding)

① filter size 3x3 5x5

|    |   |    |
|----|---|----|
| 1  | 0 | 1  |
| 0  | 0 | 0  |
| -1 | 0 | -1 |

3x3



|   |   |   |   |   |   |
|---|---|---|---|---|---|
| 4 | 9 | 2 | 5 | 8 | 3 |
| 5 | 6 | 2 | 4 | 0 | 3 |
| 2 | 4 | 5 | 4 | 5 | 2 |
| 5 | 6 | 5 | 4 | 7 | 8 |
| 5 | 7 | 7 | 9 | 2 | 1 |
| 5 | 8 | 5 | 3 | 8 | 4 |

+

|   |   |    |
|---|---|----|
| 1 | 0 | 1  |
| 1 | 0 | -1 |
| 1 | 0 | -1 |

=

|   |  |  |  |
|---|--|--|--|
| 2 |  |  |  |
|   |  |  |  |
|   |  |  |  |
|   |  |  |  |

parameters  
 size  $f=3$   
 stride  $s=1$   
 padding  $p=0$

$n_H \times n_W = 6 \times 6$

$2 = 4 \times 1 + 9 \times 0 + 2 \times -1 + 1 \times 5 + 6 \times 0 + 2 \times -1 + 2 \times 1 + 4 \times 0 + 5 \times -1$

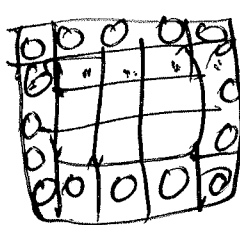
kernel size is  $3 \times 3$

② window slide 

|   |   |  |
|---|---|--|
| 2 | 6 |  |
|   |   |  |
|   |   |  |

stride 1  $\rightarrow$  shift by 1  
 relate  $ip\ size \Rightarrow o/p\ size$

$n_H' = n_H - f + 1$   
 $6 - 3 + 1 = 4 \times 4$

padding with zero padding 

$p=1$

$ip\ 3 \times 3 \rightarrow o/p\ 5 \times 5$

$p=2$  two zeros

reflect padding another instead zero replicating (2)

$$(n - f + 1) = n_H$$

$$(n - f + 2p + 1) = n_H$$

$$p = \frac{f-1}{2}$$

3x3  $\rightarrow$   
 $\hookrightarrow 5 \times 5$   
 2 times so (2p)

Same longer (A)

Pad zero. (Why) or else reduce size and increase

|   |   |   |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 4 | 9 |
| 0 | 5 | 6 |

$\rightarrow -15$

contribute from neighbors also

$$n - f + 2p + 1 = n_H$$

$$n_H - f + 2p + 1 = n_H$$

$$p = \frac{f-1}{2}$$

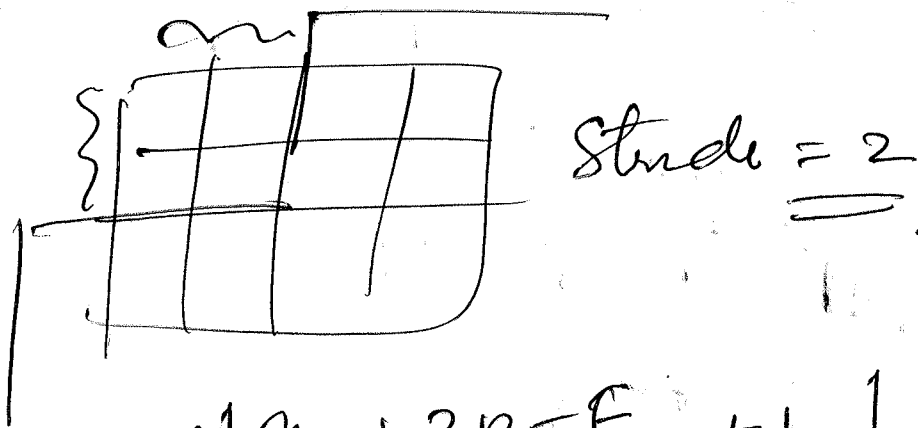
eg

$$(6 \times 6) (3 \times 3)$$

$$= 4 \times 4$$

padding = valid (zero padding)  
 = "same" (o/p is same)  
 tip some size

Stride  
 $S = 2$  shift filter by 2



$$n_A = \left\lfloor \frac{n_H + 2p - F}{S} + 1 \right\rfloor$$

how give an example

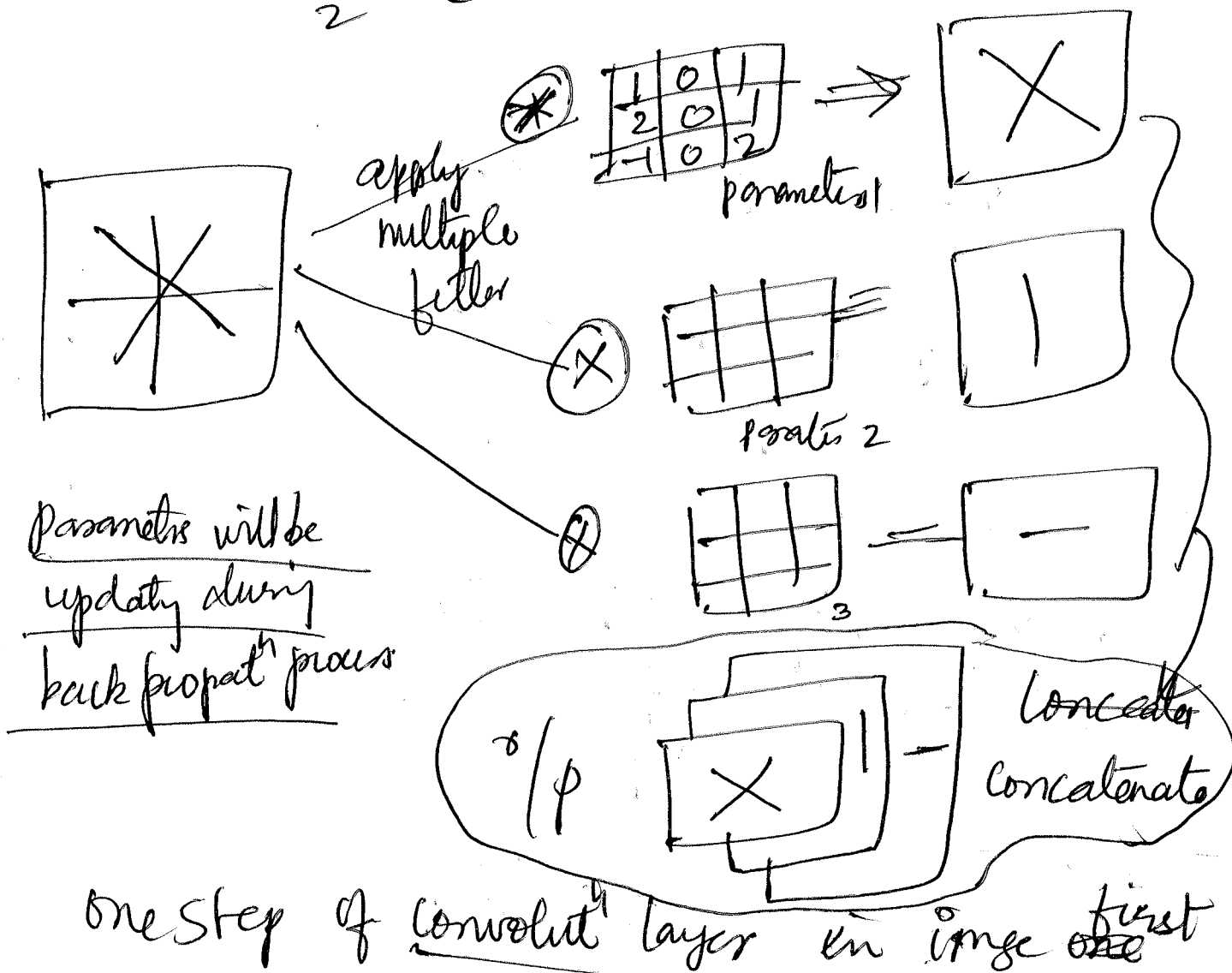
img 6x6  
filter 3x3  
S=2

p=0

$$\frac{6 - 0 - 3}{2} + 1$$

2x2

$$\frac{3}{2} = \lfloor 1.5 + 1 \rfloor = \lfloor 2.5 \rfloor = 2$$

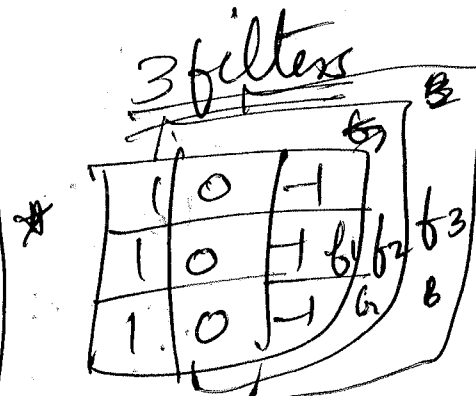
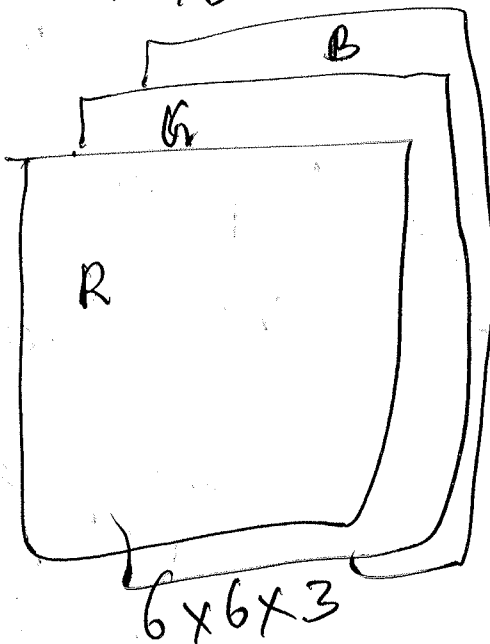


Initial wt random then get optimized  
 which will <sup>converge with some</sup> ~~be~~ good features  
 and that will get detected

gray scale image and passed to second layer

next Red color image

RGB

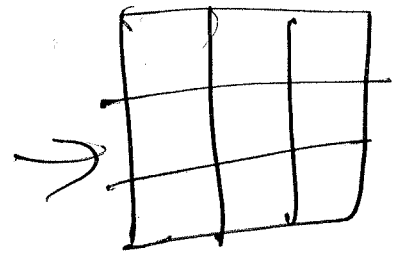


$$S_{yc} = F = 3$$

$$\text{Channel } n_c = 3$$

$$\text{Stride } S = 1$$

$$\text{Padding } P = 0$$



2D image

$$2 = \text{Red} \times \text{filter} + \text{Green} \times \text{filter} + \text{Blue} \times \text{filter}$$

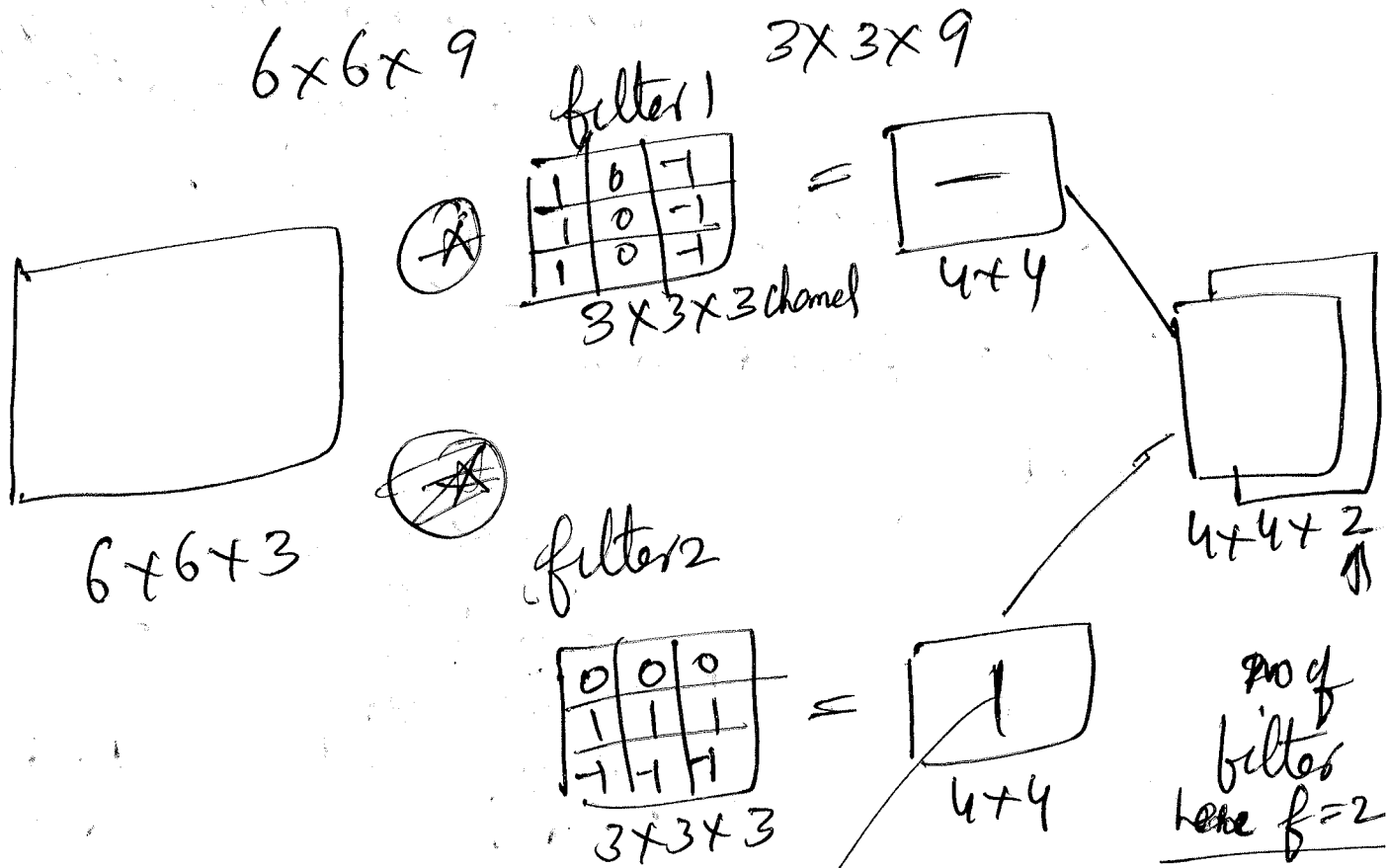
Stride

$$3D \rightarrow 2D$$

3D convolve with 3 filter  $\rightarrow$  2D image  
 3 channel filter

$$6 \times 6 \times 3 \text{ (*) } 3 \times 3 \times 3 \rightarrow 2D \text{ image single channel}$$

if img is hyperspatial 9 channels



horizontal  
vertical  
diagonal filters

Convolut<sup>n</sup> of multiple filters

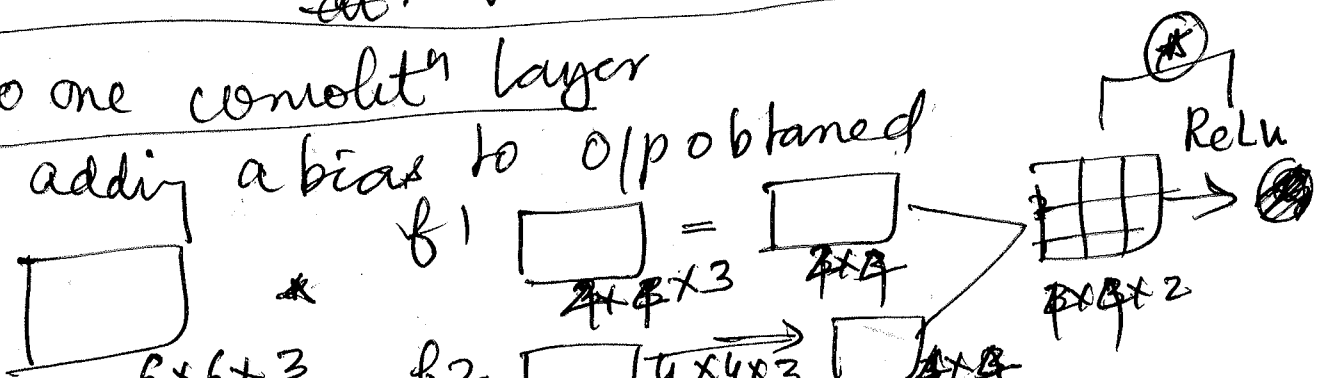
over RGB

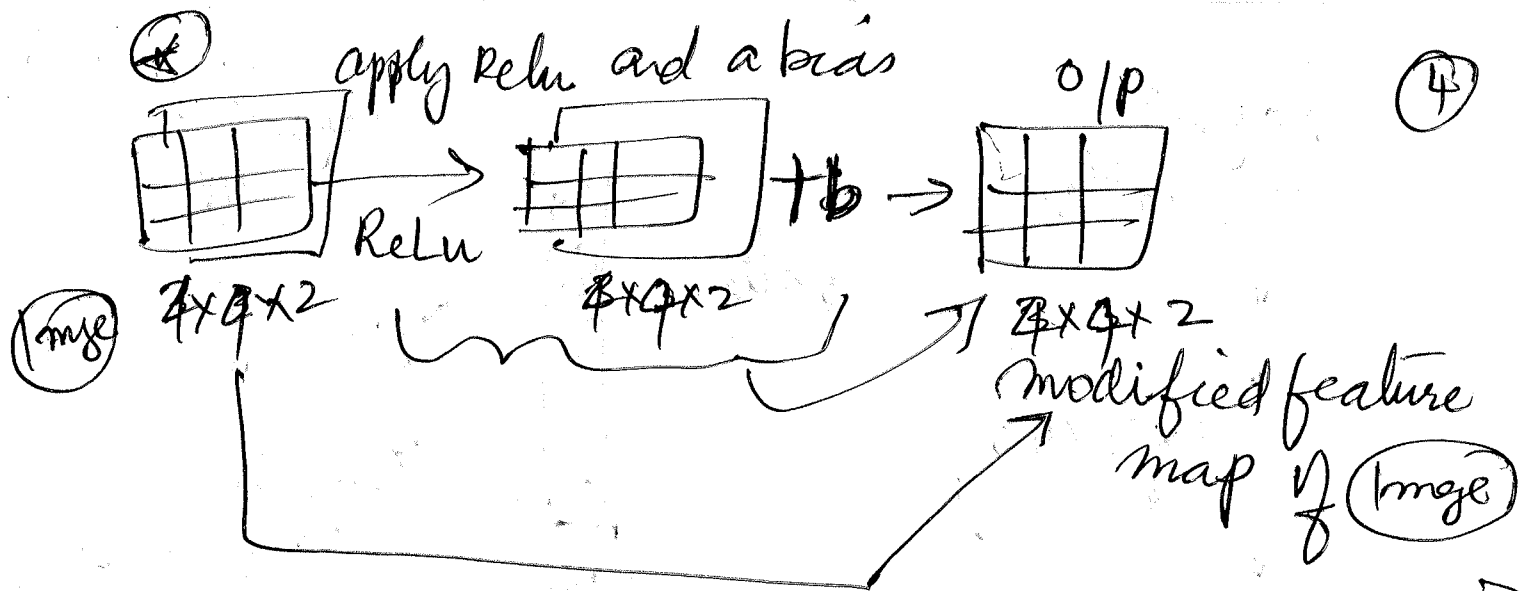
So may detect horizontal line or  
di. vertical line

one filter  
will have  
3 volume

So one convolut<sup>n</sup> layer

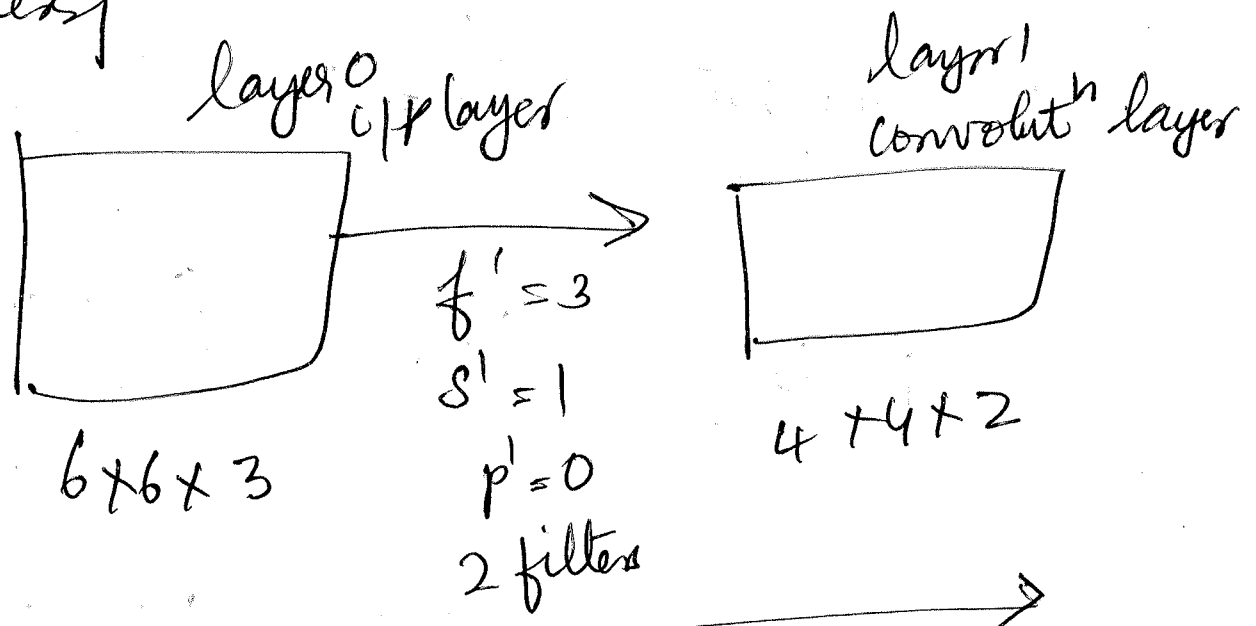
adding a bias to o/p obtained



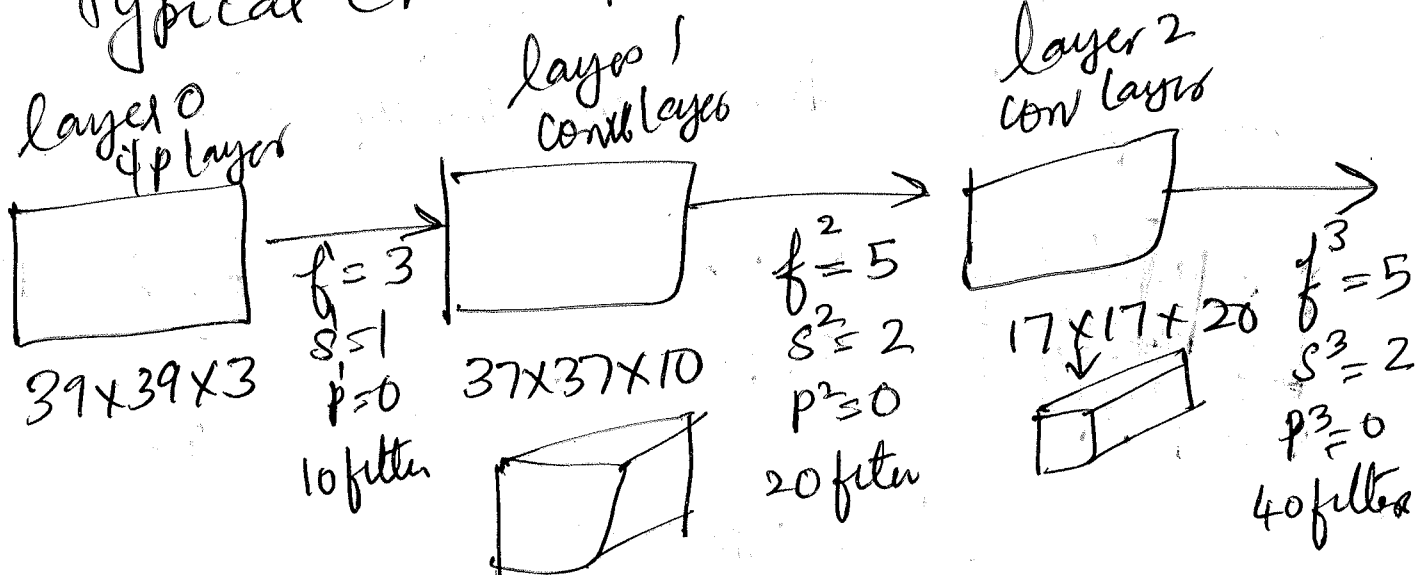


o/p will be passed to next layer for further

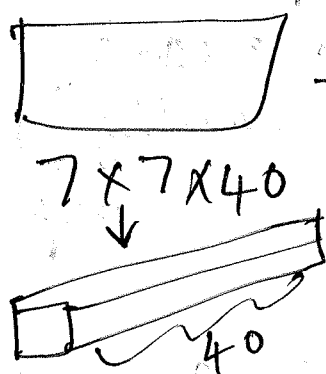
process;



typical CNN w/w



Layer 3  
convol<sup>n</sup> layer

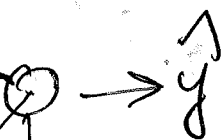


flatten



1960  
array is FC

Layer 4  
O/P



logistic /

Softmax

1 node

defects

(1960, 1)

or applicat<sup>n</sup>

$$\left\lceil \frac{39 + 0 - 3}{1} + 1 \right\rceil = 37$$

padding = valid.

$$\left\lceil \frac{37 + 0 - 5}{2} + 1 \right\rceil = 17$$

pooling layer  $\rightarrow$  to reduce the dimensionality of image  
max pooling  $\&$  AV dimensionality = 7

|   |   |   |   |
|---|---|---|---|
| 4 | 9 | 2 | 5 |
| 5 | 6 | 2 | 4 |
| 3 | 4 | 5 | 4 |
| 5 | 6 | 8 | 4 |

|   |   |
|---|---|
| 9 | 5 |
| 8 | 8 |

$f = 2$   
 $s = 2$

av  $\rightarrow$

|     |     |
|-----|-----|
| 6.0 | 3.3 |
| 4.3 | 5.3 |

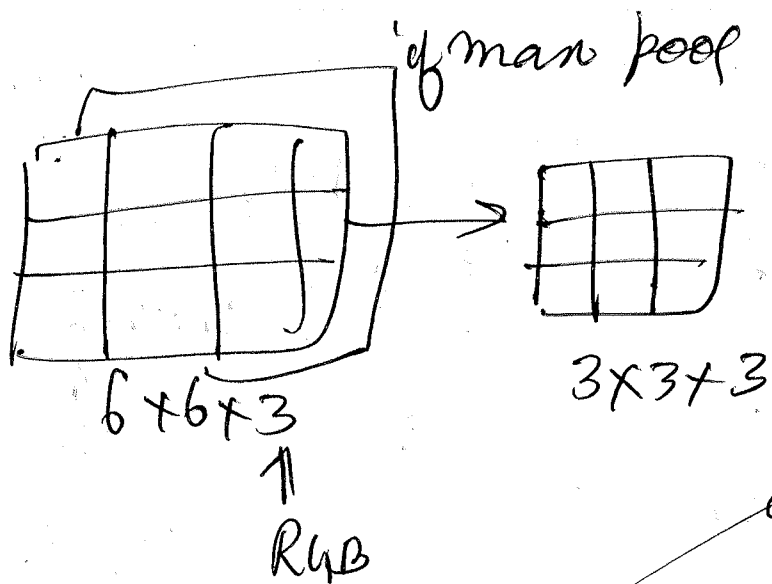
$$\frac{4 - f - 2p + 1}{s}$$

$$\frac{4 - 2 - 0 + 1}{2}$$

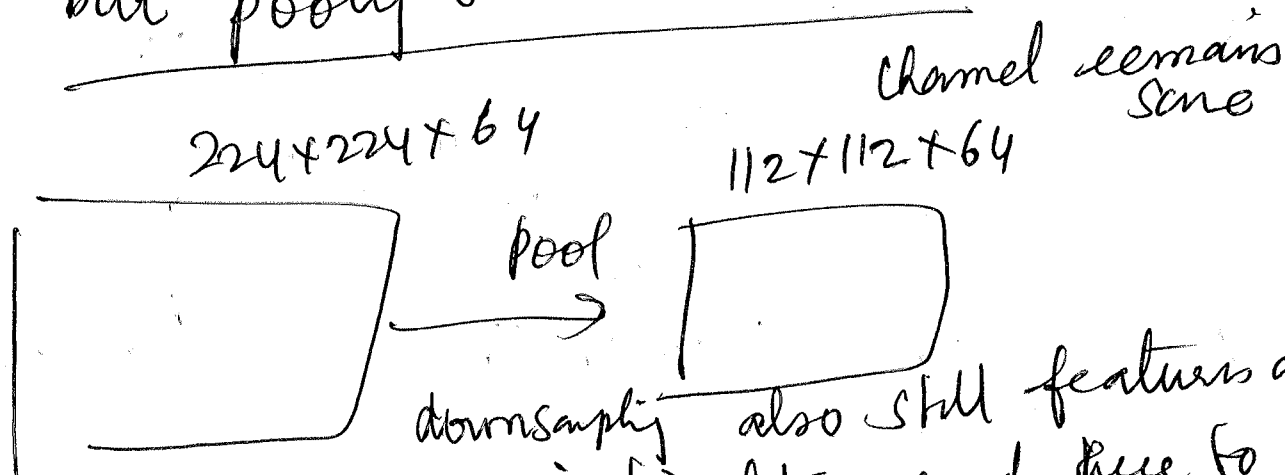
$$= 2$$



⑤



down sampling  
In convnet<sup>n</sup> we use 3 channel and add  
but pooling on each channel.



only reduce dimensionality  
also still features are  
and due to  
interpret not  
like conv layer  
reduces size

(Saving Space and computational time)  
we can interpret eg edges etc

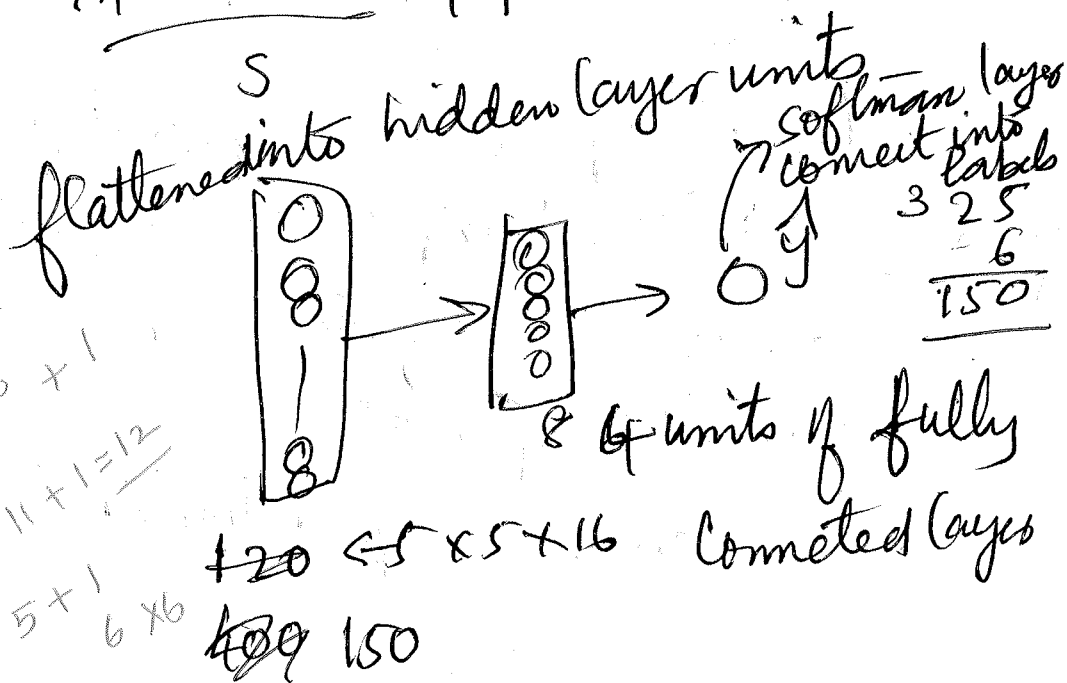
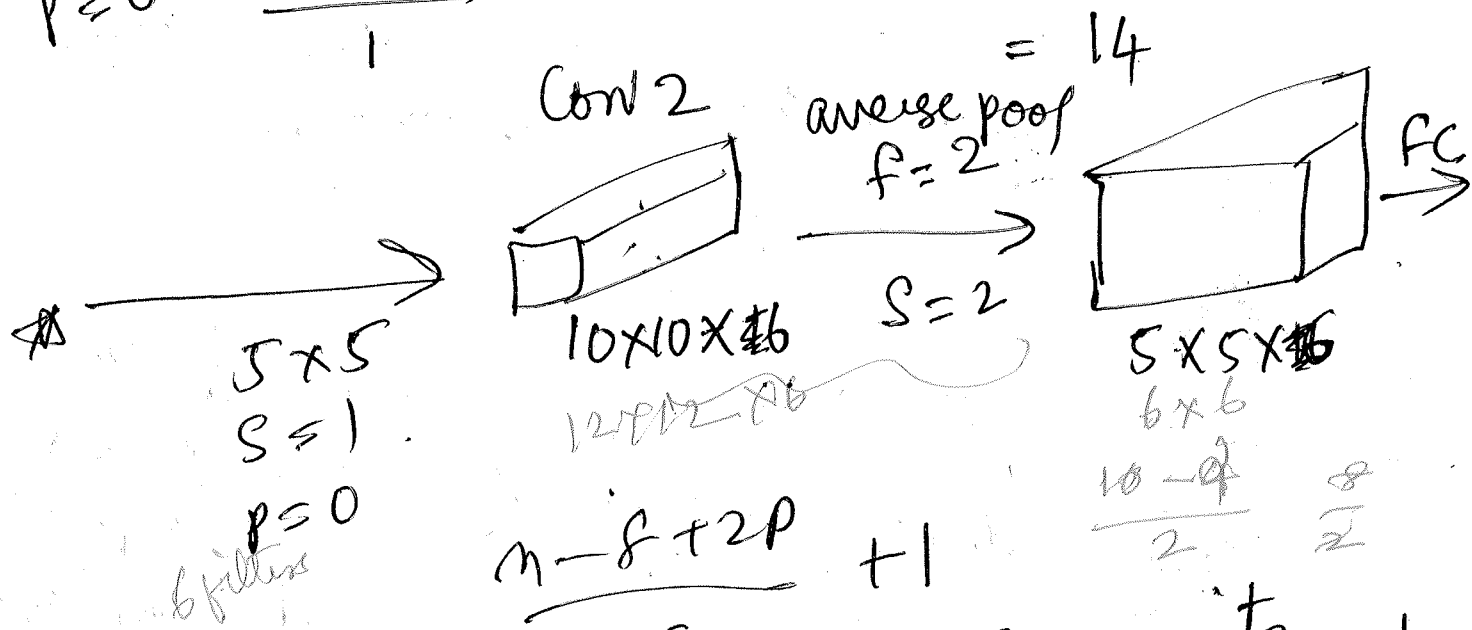
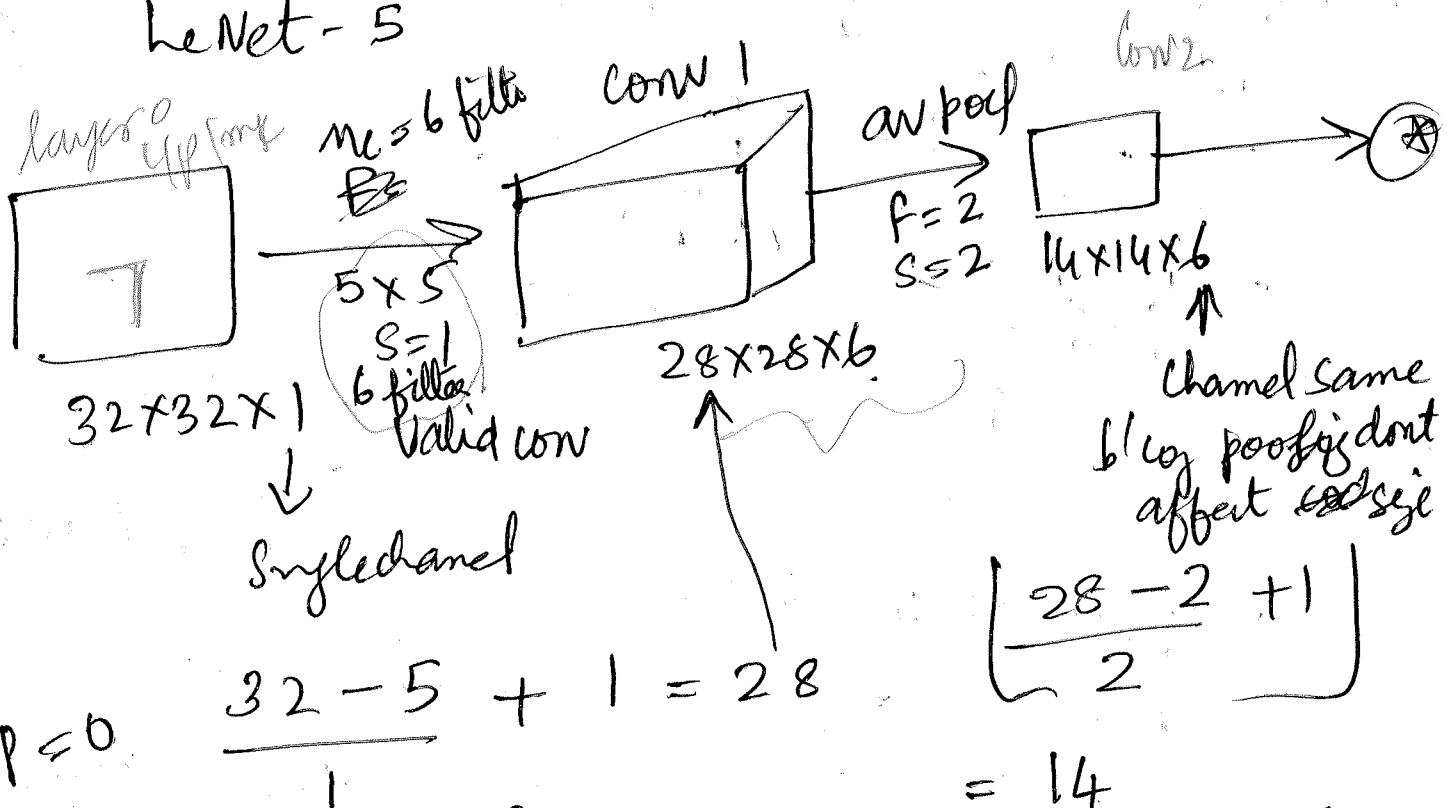
padding

- retain all features obtained after performing convolution operation
- reduces computational time

(Audi@#2026)

Audi

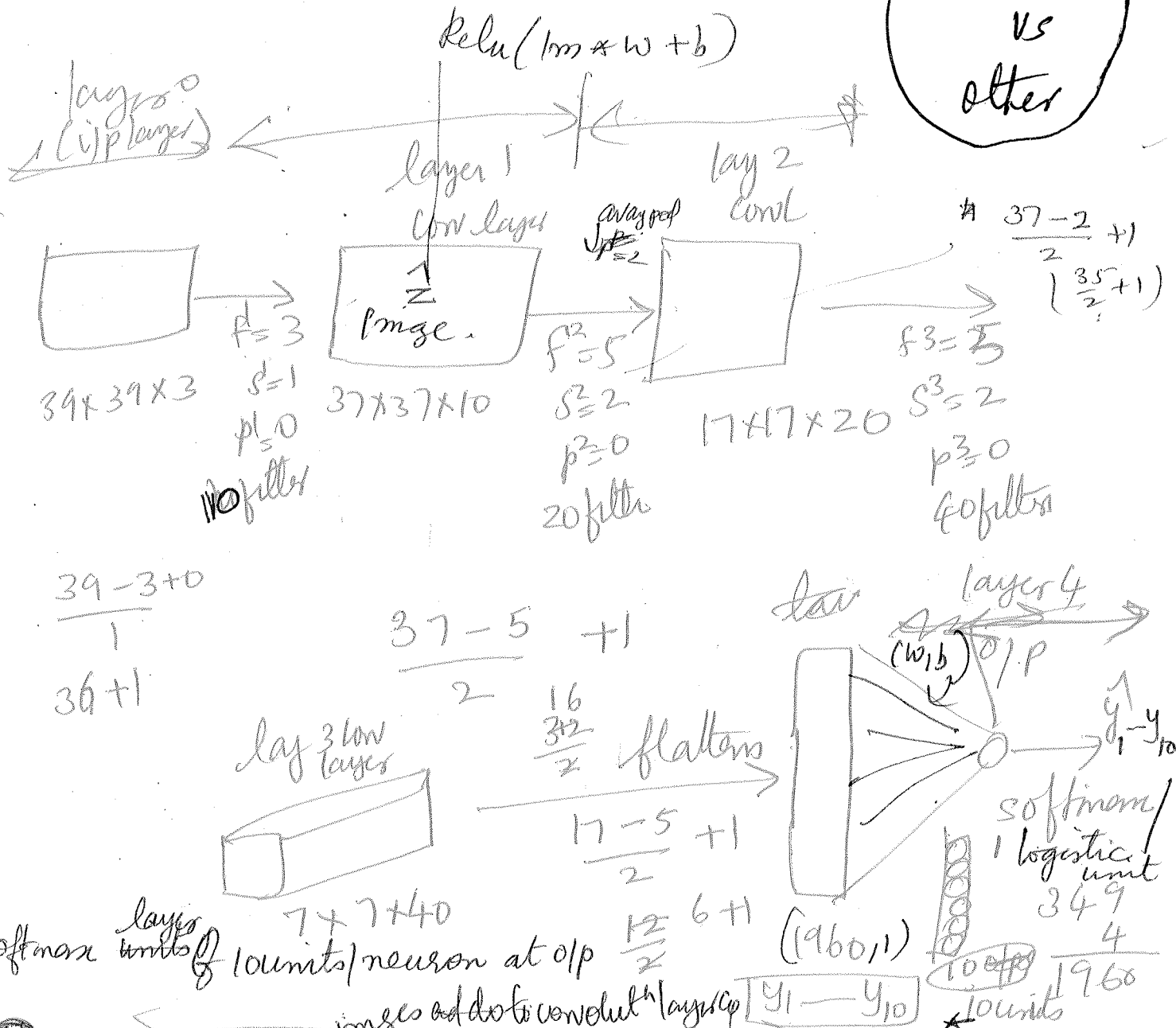
# LeNet-5



② cube size ↓ channel width ↑

① when we do back propagat<sup>n</sup> weight in filters get updated but not in pooling layer filter.  
 not involved in BP.

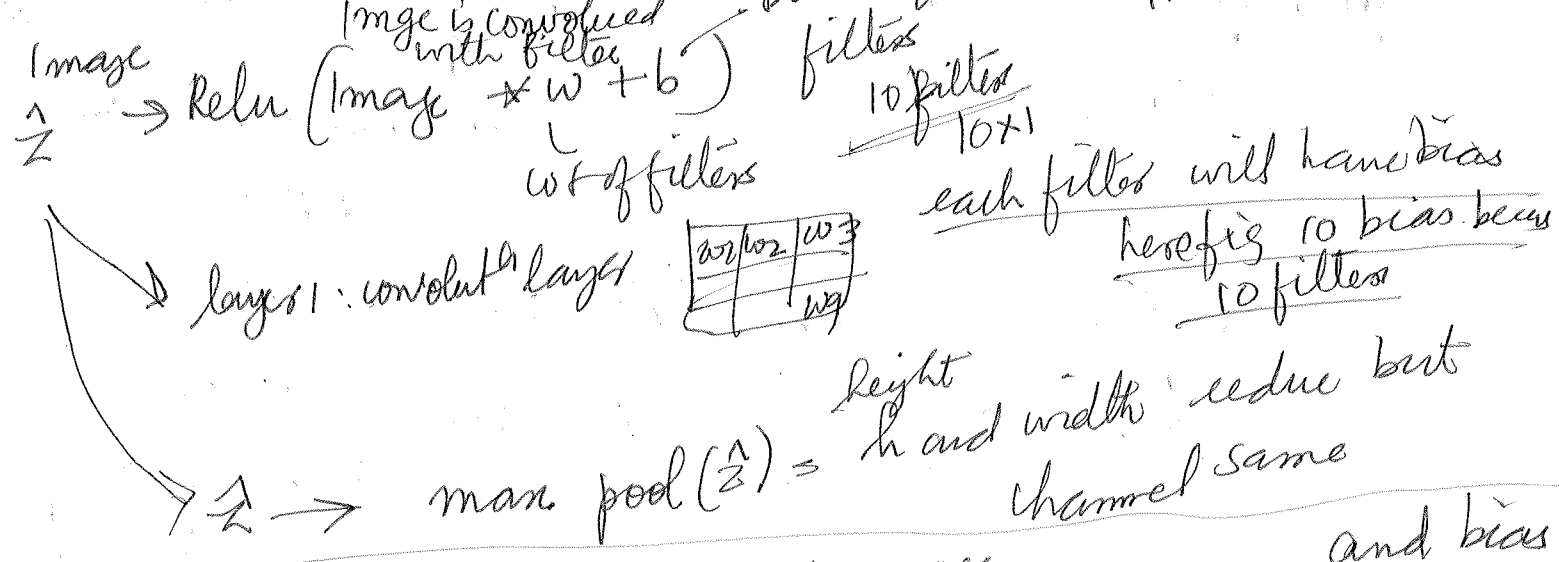
Softmax (multiclass) → 0 or 1  
 Sigmoid (multi label) → only one class vs other



→ Softmax units of 10 units/neuron at o/p

① feeding the i/p, layers do transform<sup>n</sup> is the process of forward propagat<sup>n</sup>  
 binary class 0/1  
 multiple classify class

# binary o/p only one neuron (0 or 1) depends on express mathematically



backward propact<sup>n</sup> - purpose for updating of weights of filters

wt of pooling filters wont be updated

find errors & then minimize loss

1. cal diff of vectors in terms of

Mean absolute error  $\leftarrow L_1$  loss or  $L_2$  loss  $L_2 = \frac{1}{2} \|y - \hat{y}\|^2$  mean square error

2. then backpropagate the loss

predicted loss

original o/p

so that we get these o/p

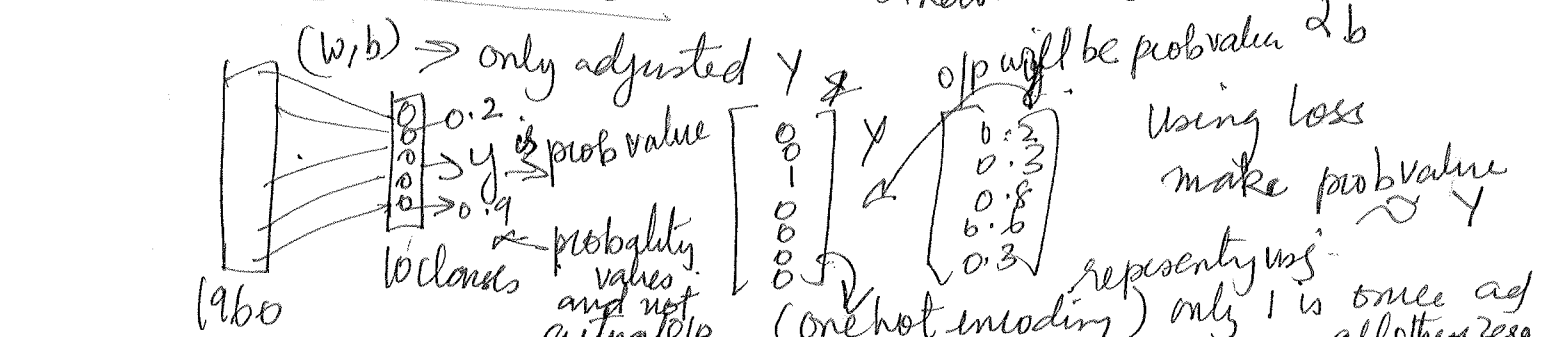
partial derivative of loss w.r to w & b

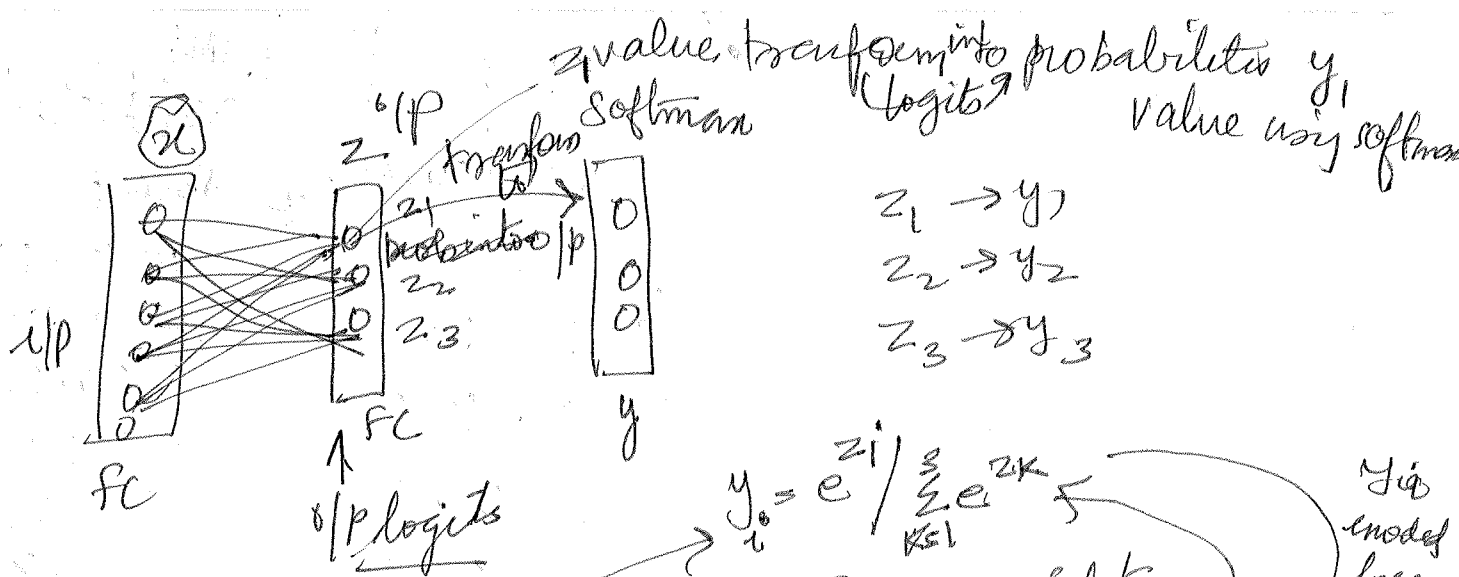
$\frac{\partial L}{\partial w}$ ,  $\frac{\partial L}{\partial b}$

update weights using gradient descent

$w_{\text{new}} = w_{\text{old}} - \alpha \frac{\partial L}{\partial w}$

$b_{\text{new}} = b_{\text{old}} - \alpha \frac{\partial L}{\partial b}$





Equation for the Softmax function:

$$y_i = \frac{e^{z_i}}{\sum_{k=1}^3 e^{z_k}}$$

Annotations include:

- "logits  $\rightarrow$  probability value"
- "3 (class)"
- "loss =  $-\sum_{i=1}^3 y_i \log(y_i)$ "
- "labels one hot encoded vector"
- "this is for 3 class"
- "y is encoded loss"
- "y is encoded loss"

$z$  is of units or logits

\* Cross entropy loss

②  $y_i$  encoded into log and gives you loss

Generalizes this multiple class. 3 class.

Equation for the Cross Entropy Loss:

$$L = -\sum_{i=1}^3 y_i \log(y_i)$$

Annotations include:

- "cross entropy loss"
- "loss"

Equation for the Softmax function:

$$y_i = \frac{e^{z_i}}{\sum_{k=1}^3 e^{z_k}}$$

Annotations include:

- "(probability value)"
- "will get from here"
- "next"
- " $z_i = \sum_{j=1}^n w_{ji} x_j + b$ "

"Softmax transfer"

next of nodes in the layer (x)

(f.c. version of neural net w/ with w & b bias)

CSE DS

24/7/2026

Present

10, 24, 25, 45, 36

49, 39, 40, 37, 28

9

CSE DS

22/7/2026

Present

49, 39, 32

54, 24, 10, 5

52, 40, 37

29, 24, 36

41, 45, 27

28, 46, 19

41, 31, 56

7

AIML

22/7/2026

absent

2, 10, 12, 15,

18, 19, 20, 21,

22, 24, 25,

29, 31, 32

33, 35, 36

37, 340

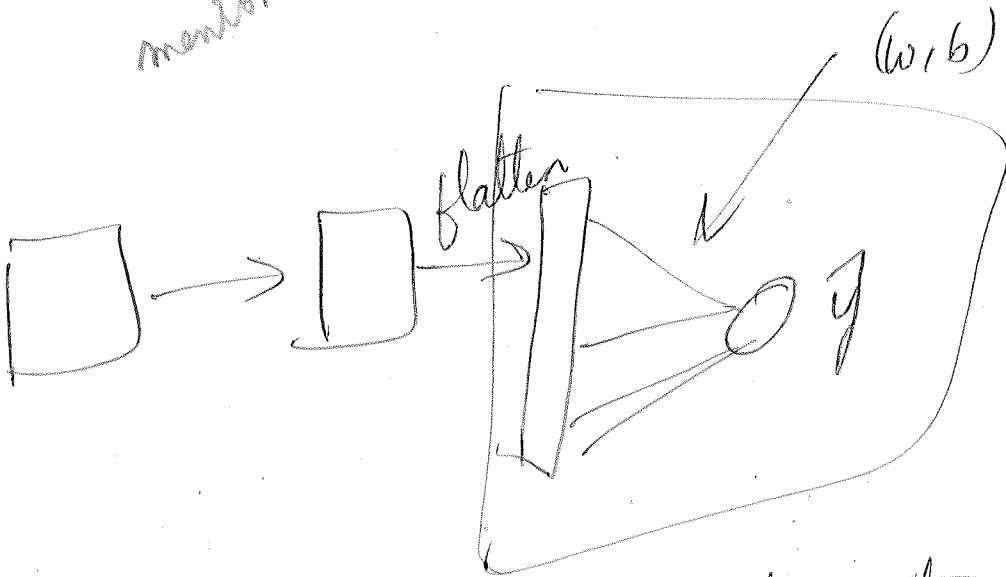
41, 42, 4

48, 49,

54, 55, 56

52, 58

mentor @ [kaggle.org](https://www.kaggle.com)



one hot encoding - convert categorical data (labels)  
into binary vector where one element is 1  
other zero

Cat = 1

Cat = 2

then = 3 ~

→ Categorical labels

attendance - fill  
→ send to all faculty  
sick leave

→ Afal, word

→ Projects (misc project)

→ lab prediction